Student: Xirong Zhang **Date:** 11/30/25	**Instructor:** Dmitriy Ivanov **Course:** Math 120 Fall 2025	**Assignment:** Section 1.1

1. Solve the system by using elementary row operations on the equations. Follow the systematic elimination procedure.

$$x_1 + 5x_2 = 5$$
$$4x_1 + 7x_2 = -6$$

Find the solution to the system of equations.

$(x_1, x_2) =$ $(-5, 2)$
(Simplify your answer. Type an ordered pair.)

2. Consider the accompanying matrix as the augmented matrix of a linear system. State in words the next two elementary row operations that should be performed in the process of solving the system.

$$\begin{bmatrix} 1 & -5 & 5 & 0 & -2 \\ 0 & 2 & -6 & 0 & 5 \\ 0 & 0 & 1 & 4 & -3 \\ 0 & 0 & 3 & 10 & 1 \end{bmatrix}$$

What should be the first elementary row operation performed? Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- **A.** Replace row 4 by its sum with -3 times row 3.
  (Type an integer or a simplified fraction.)
- **B.** Replace row 2 by its sum with _____ times row 4.
  (Type an integer or a simplified fraction.)
- **C.** Interchange row 3 and row 2.
- **D.** Scale row 1 by _____.
  (Type an integer or a simplified fraction.)

What should be the second elementary row operation performed? Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- **A.** Replace row 3 by its sum with _____ times row 2.
  (Type an integer or a simplified fraction.)
- **B.** Scale row 4 by $-\frac{1}{2}$.
  (Type an integer or a simplified fraction.)
- **C.** Replace row 1 by its sum with _____ times row 4.
  (Type an integer or a simplified fraction.)
- **D.** Interchange row 1 and row 2.

3. The augmented matrix of a linear system has been reduced by row operations to the form shown. Continue the appropriate row operations and describe the solution set of the original system.

$$\begin{bmatrix} 1 & 8 & 3 & -3 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 1 & 3 \\ 0 & 1 & -1 & 2 \end{bmatrix}$$

Select the correct choice below and, if necessary, fill in the answer boxes to complete your choice.

- A. The solution set has exactly one element, (, ,).
  (Type integers or simplified fractions.)
- B. The solution set has infintely many elements.
- C. The solution set is empty.

4. The augmented matrix of a linear system has been reduced by row operations to the form shown. Continue the appropriate row operations and describe the solution set of the original system.

$$\begin{bmatrix} 1 & -4 & 6 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 4 & 0 \end{bmatrix}$$

Select the correct choice below and, if necessary, fill in the answer boxes to complete your choice.

- A. The solution set contains one solution, (0 , 0 , 0).
  (Type integers or simplified fractions.)
- B. The solution set has infinitely many elements.
- C. The solution set is empty.

5. The augmented matrix of a linear system has been reduced by row operations to the form shown. Continue the appropriate row operations and describe the solution set of the original system.

$$\begin{bmatrix} 1 & -1 & 0 & 0 & -5 \\ 0 & 1 & -3 & 0 & -6 \\ 0 & 0 & 1 & -2 & 2 \\ 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

Select the correct choice below and, if necessary, fill in the answer boxes to complete your choice.

- A. The solution set contains one solution: (19 , 24 , 10 , 4).
  (Type integers or simplified fractions.)
- B. The solution set is empty.
- C. The solution set has infinitely many elements.

6. Determine if the given system is consistent. Do not completely solve the system.

$$\begin{aligned} 2x_1 \quad\quad + 6x_3 \quad\quad &= 10 \\ x_2 \quad\quad - 4x_4 &= 4 \\ -5x_2 + 6x_3 + x_4 &= 5 \\ 6x_1 \quad\quad\quad + 8x_4 &= -1 \end{aligned}$$

Choose the correct answer below.

- A. The system is consistent because the system can be reduced to a triangular form that indicates that no solutions exist.
- B. The system is inconsistent because the system cannot be reduced to a triangular form.
- C. The system is consistent because the system can be reduced to a triangular form that indicates that a solution exists.
- D. The system is inconsistent because the system can be reduced to a triangular form that contains a contradiction.

7. Determine whether the statement below is true or false. Justify the answer.

Every elementary row operation is reversible.

Choose the correct answer below.

- **A.** The statement is false. The only reversible row operation is interchanging two rows.
- **B.** The statement is false. The only reversible row operations are scaling a row and interchanging two rows.
- **C.** The statement is true. Interchanging can be reversed by scaling, and scaling can be reversed by replacement.
- **D.** The statement is true. Replacement, interchanging, and scaling are all reversible.

---

8. Determine whether the statement below is true or false. Justify the answer.

A 5×6 matrix has six rows.

Choose the correct answer below.

- **A.** The statement is false. A 5×6 matrix has five rows and five columns.
- **B.** The statement is false. A 5×6 matrix has five rows and six columns.
- **C.** The statement is true. A 5×6 matrix has five columns and six rows.
- **D.** The statement is true. A 5×6 matrix has six columns and six rows.